Jiajie Zhang

Mobile Autonomous Robotic Systems(MARS) Lab , ShanghaiTech University

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Experience

AI R&D Center, Central Research Institute, Wolong Electric — **July 2025 - Oct. 2025**
Research Assistant — Shanghai, China

- Supervised by Alexander Kleiner

Education

ShanghaiTech University — **Sep 2023 - Present**
Computer Science and Technology — Master of Science
3.51/4.0 GPA

- Selected Courses: Robotics, Mobile Manipulation, Deep Learning
- Supervised by Professor Soeren Schwertfeger

Zhengzhou University — **Sep 2019 - Jun. 2023**
Automation — Bachelor of Engineering
3.66/4.0 GPA

Selected Publications

From Observation to Action: Latent Action-based Primitive Segmentation for VLA Pre-training in Industrial Settings — **Nov. 2025**
Under review for CVPR2026
https://jiajiezhang7.github.io/latent-action-primitive-segmenter/

Jiajie Zhang†, Sören Schwertfeger, and Alexander Kleiner

Intelligent LiDAR Navigation: Leveraging External Information and Semantic Maps with LLM as Copilot — **Mar. 2025**
Accepted by IROS 2025
https://arxiv.org/abs/2409.08493

Fujing Xie†, **Jiajie Zhang** and Sören Schwertfeger

Selected Projects

AGLoc++: WiFi-Fused Global Localization and Monte Carlo Enhanced Tracking in Hierarchical Area Graph — **Nov. 2024 - Present**
https://jiajiezhang7.github.io/projects/1_agloc_project/

- **Led AGLoc++ development**: Ported Area Graph LiDAR localization system to ROS2, integrated with the Nav2 stack, implemented WiFi-aided kidnap recovery, developed odometry-fused Monte Carlo tracking with advanced re-localization.
- **Enhance Robustness:** implement techniques such as clutter filtering, architectural matching, weighted ICP, corridor-aware downsampling,ensuring robust, high-precision localization in dynamic indoor environments.

Languages

English
TOEFL iBT: 106 (Reading: 30/30, Writing: 27/30, Listening: 27/30, Speaking: 22/30)

Skills

Python & C++
Object-Oriented Programming (OOP), NumPy, PyTorch, Matplotlib / Encapsulation, Inheritance, Polymorphism

Tools
ROS/ROS2, Navigation2, PyTorch, Linux, Git, LaTex, Cursor, Windsurf